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Aim

To design and implement an Object-Oriented Student Information System using Java. The program demonstrates core OOP concepts such as classes, objects, encapsulation, constructors, and data abstraction. It allows users to input student details and display them interactively.

Procedure

Step 1: Defining the Students Class

- A Students class was created with private data members such as name, age, roll number, and contact number.
- Encapsulation was achieved using private variables and public setter/getter methods.

Step 2: Implementing Setters and Getters

- Setter methods were used to assign values to private variables.
- Getter methods were used to retrieve values for display.

Step 3: Display Method

- A display() method was implemented to print all student details in a structured format

Step 4: Designing the Main Class (Solution)

- The Solution class contains the main() method.
- It uses the Scanner class to accept user input for student details.
- An object of the Students class was created, and setter methods were used to store the input values.

Step 5: Applying Object-Oriented Concepts

- **Encapsulation:** Private data members with public setters/getters.
- **Abstraction:** User interacts only through methods without accessing internal variables directly.
- **Object Interaction:** Solution class interacts with Students class.

- **Reusability:** The Students class can be reused for multiple student records

Step 6: Executing the Main Class

- The program execution starts in the main() method.
- User inputs student details, which are stored and displayed using the Students class methods.

Step 7: Testing The program was tested by:

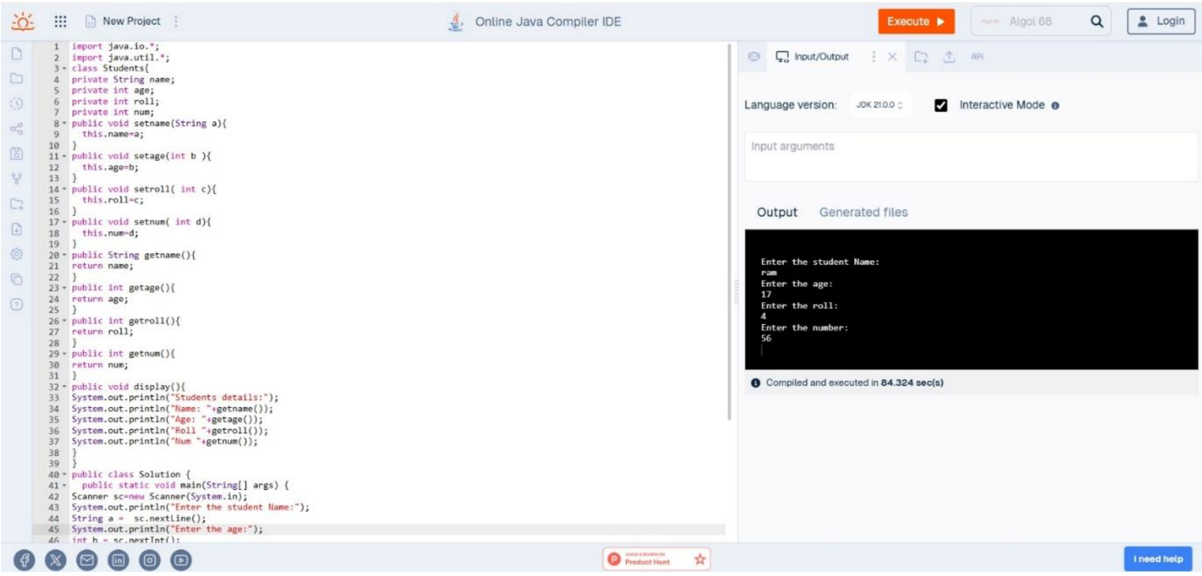
- Entering different student details.
- Displaying the stored details correctly.
- Handling invalid inputs gracefully.

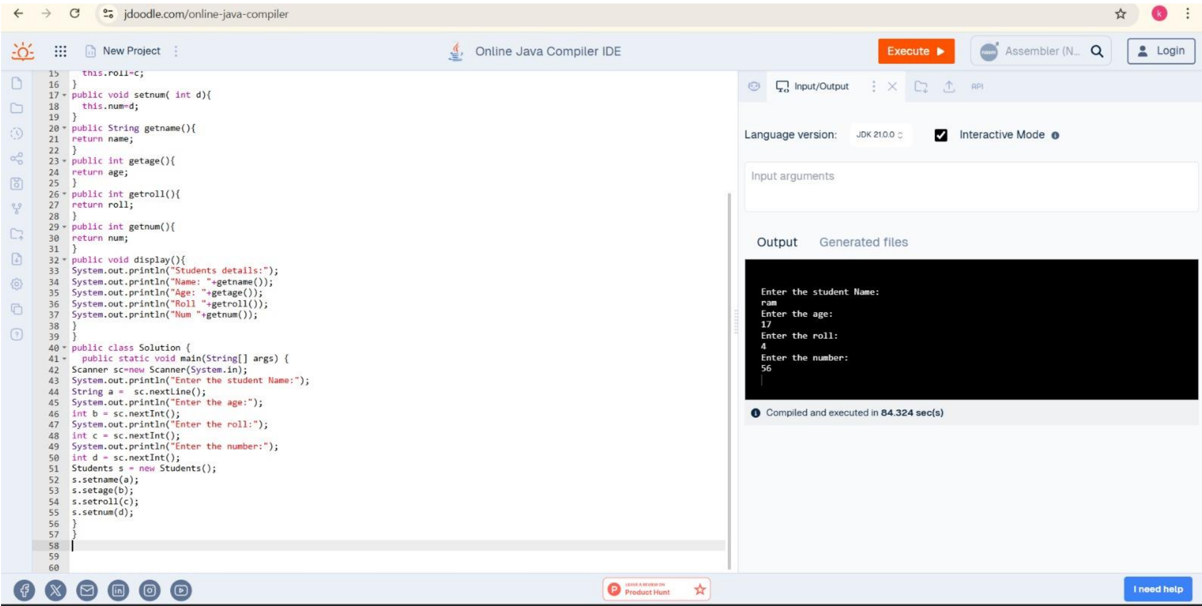
CODE

```
import java.io.*;

import java.util.*;

class Students{

private String name;

private int age;

private int roll;

private int num;

public void setname(String a){

  this.name=a;

}

public void setage(int b ){

  this.age=b;

}

public void setroll( int c){

  this.roll=c;

}

public void setnum( int d){

  this.num=d;

}
```

```
public String getname(){
return name;
}
public int getage(){
return age;
}
public int getroll(){
return roll;
}
public int getnum(){
return num;
}
public void display(){
System.out.println("Students details:");
System.out.println("Name: "+getname());
System.out.println("Age: "+getage());
System.out.println("Roll "+getroll());
System.out.println("Num "+getnum());
}
}
public class Solution {
  public static void main(String[] args) {
Scanner sc=new Scanner(System.in);
System.out.println("Enter the student Name:");
String a =  sc.nextLine();
System.out.println("Enter the age:");
int b = sc.nextInt();
System.out.println("Enter the roll:");
int c = sc.nextInt();
```

```
System.out.println("Enter the number:");
int d = sc.nextInt();
Students s = new Students();
s.setname(a);
s.setage(b);
s.setroll(c);
s.setnum(d);
}
}
```





Result

The Student Information System was successfully implemented using Object-Oriented Programming in Java. The program allows users to input and display student details accurately. All OOP principles such as encapsulation, abstraction, and object interaction were effectively demonstrated. The system produced correct outputs and handled user input efficiently.